James Bao

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Professional Summary

*I am dedicated to **improving human lives** through a **beautiful combination** of hardware and firmware.*
I am **hungry for best practises**, with a commitment to **continuous learning** and personal growth.
I have a **proven problem-solving** mindset, and leave things **better than I found them**.
I am an effective **communicator & collaborator**, leading multiple **top university design teams** to success.
I am an **asset to any team**—I don't make promises I cannot keep.

Education

Bachelor of Engineering (Honours)

2021 – 2024

The University of Auckland

- Part IV **Computer Systems Engineering**
- Student Representative and **Teaching Assistant**
- Cumulative GPA of 8.4/9.0 (**A** – **A+**)
- Dean's Honours List (top 5%)

Technical Skills

TypeScript/JavaScript

- Front-end **browser** and back-end **Node.js** applications
- Object-oriented and functional patterns
- **async/await**, Promises, and callbacks
- **Serverless functions** with Next.js and Cloudflare Workers

Next.js/React

- Next.js Pages and **App Router**
- React **custom hooks**
- Props and state management
- API routes and **server/client components**
- Lazy-loading and memoisation
- **CSS Modules**, Sass, and **Tailwind CSS**

Git/GitHub

- Fluency with **feature branches**, pull requests, issues, and GitHub Actions
- **7800+ contributions** accrued over the past **6 years**

Other Languages I also comfortable with **C/C++**, and have familiarity with **C#**, **Python**, **Java**, and **MATLAB**.

Industry Experience

Firmware Engineer Intern

Nov 2023 – Feb 2024

EROAD Ltd — *Auckland, NZ*

- Writing **C++ RTOS application** service with **GoogleTest** unit & component tests
- Created a **Next.js/React** front-end dashboard application with **Zod** data validation
- Developed a 'highly polished internal support tool' that **reduced 4+ hours**-worth of work to **less than 2 minutes**
- Implemented **REST API** business logic to facilitate asynchronous **remote file retrieval** from in-field devices
- Produced **security architecture**, threat modelling, and data flow diagrams to identify & secure attack vectors

Embedded Software Engineer Intern

Nov 2022 – Mar 2023

Windcave Ltd — *Auckland, NZ*

- Writing **C firmware** on STM32U585 **ARM** board
- Performed **rapid prototyping** of external **QSPI and OSPI** flash memories to inform hardware revision
- Implemented a **FreeRTOS** graphical Point-of-Sale system prototype on a Windcave payment terminal to satisfy **long-outstanding customer demand**

Freelance Software Developer

2020 – 2021

- Back-end development with JavaScript, **Node.js**, and **MongoDB**
- Built a custom Discord bot tailored to streamline **digital order management** and processing

Project Experience

notion-assignment-import

2022

jamesnzl.xyz/notion-assignment-import

- **Cross-browser** extension that loads Canvas assignments into a Notion database
- **5200+ unique users** in **eighteen months**
- Variety of tools including **Gulp**, Browserify & ESBuild, **Sass**, and PostCSS
- Learnt more about **REST APIs**, HTTP and HTTP clients, and **OAuth2**

- Git and GitHub
- TypeScript, HTML
- Sass and PostCSS
- Gulp, Browserify, and ESBuild
- Insomnia

Pronto

2020 – 2021

aucklandarmycadets.github.io/Pronto — *City of Auckland Cadet Unit*

- Full-featured Discord bot to **streamline** Unit **procedures**
- Comprehensive **codebase documentation** with **JSDoc**
- Experimenting with **Cloud Computing** services including DigitalOcean, Linode, Vultr, and currently **Oracle Cloud**

- JavaScript, Node.js
- MongoDB
- JSDoc
- Virtual Private Servers
- SSH, Bash, Unix CLI

Unit Website

2022

github.com/aucklandarmycadets/aucklandarmycadets — *City of Auckland Cadet Unit*

- A **modern, performant website** to overhaul my Cadet Unit's online presence
- Serverless functions with both **Next.js** and **Cloudflare Workers**
- Created my **own CDN** to cache Notion's ephemeral images and **optimise performance**
- Experience in managing **DNS records**, administering Google Search Console, and enhancing **SEO**

- Next.js (and React) with Vercel
- Serverless functions
- DNS Records, SEO

Other Projects More projects can be found on my **GitHub profile**.

Achievements

Dean's Honours List

2022, 2023

auckland.ac.nz — *Faculty of Engineering*

Awarded to students who are either placed in the top 5% of their Engineering study in their discipline and part, and/or have attained a GPA of at least 8.25 from their enrolled Engineering courses for that year.

First in Course Awards

2021, 2022, 2023

auckland.ac.nz — *Faculty of Engineering*

First in Course Awards are given each semester to students who have achieved the highest overall mark in a particular taught course.

- COMPSYS 301: **Hardware Firmware Systems Design**
- COMPSYS 303: **Microcomputers and Embedded Systems**
- COMPSYS 209: **Analogue and Embedded Firmware Design**
- ENGGEN 131: Introduction to **Engineering Computation and Software Development**

Other

Pre-2021

- Bayonet of Honour — *City of Auckland Cadet Unit*
- Top of Course Award — *New Zealand Cadet Forces*
- School Prefect and Deputy House Captain
- First in Level 3 Calculus
- First in Level 2 Physics
- First in Level 2 Programming

Passions

Outdoors I have a deep love for the outdoors, and find it an amazing outlet to recharge my batteries.

Teaching I love crafting intuitive explanations for difficult, technical concepts—and seeing the lightbulb moment.

Te Ao Māori I am passionate about our indigenous culture & language—I achieved Excellence in my NCEA Level 3 Te Reo Māori examinations, and maintain involvement with Te Whānau o Te Puna at Westlake Boys' High School.

Youth Development I now volunteer as an Officer Cadet in the New Zealand Cadet Forces, after 5 years as a cadet with 17 months as a senior Cadet Warrant Officer Class II.

References

References are available on request.